



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations
May 2025

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **How is generalization implemented in object oriented programming Language?**
A) Inheritance
B) Polymorphism
C) Encapsulation
D) Abstract class
2. **Software development begins with business analysis or users conceiving an application and formulating tentative requirement is**
A) Analysis
B) System Design
C) System conception
D) Class Design
3. **The diagram which show the flow of control among the processing steps of a computation is**
A) Use Case Diagram
B) Sequence diagram
C) Activity Diagram
D) Class Diagram
4. **The number of instances of one class that may relate to a single instance of an associated class is specified using**
A) Multiplicity
B) Composition
C) generalization
D) Aggregation
5. **In object oriented Model Traversing across Class Models is done using**
A) PBL construct
B) OCL construct
C) COL construct
D) OOP construct
6. **The flow of messages between objects in a specified order represented using**
A) Use Case Diagram
B) Sequence diagram
C) Activity Diagram
D) Class Diagram
7. **The event that is caused by the satisfaction of a Boolean expression is**
A) Signal Event
B) Object Event
C) Change Event
D) Time Event
8. **Promotion of something that is not an object into an object is called as**
A) Reification
B) Constraints
C) Aggregation
D) Generalization
9. **The relationship which adds incremental behavior to a use case in OOMD is**
A) include
B) Is a
C) extend
D) Nested
10. **Type of interaction modeling diagram which would be most appropriate for showing the flow of control in a complex business process is**
A) Use Case Diagram
B) Sequence Diagram
C) Flow Diagram
D) Activity Diagram
11. **In a use case diagram, an "actor" represent**
A) A system that handles a process
B) A role that interacts with the system
C) A method or function of the system
D) A class that encapsulates data

12. The following is NOT considered one of the core themes of Object-Oriented (OO) development?
- A) Abstraction B) Encapsulation
C) Inheritance D) Modularization
13. The purpose of signal generalization in state modeling
- A) To represent the transformation of signals in a system B) To allow more general signal to be interpreted by specific events
C) To model concurrency in the system D) To define how signals should be distributed across nested states
14. In a state model, what does a transition represent
- A) The change in the state of an object in response to an event B) The sequential execution of commands
C) The inheritance of state models D) The division of states into regions
15. The diagram is primarily used to model interactions between actors and the system in UML is
- A) Sequence Diagram B) Activity Diagram
C) Use Case Diagram D) Class Diagram
16. Skeletal structure of a program that must be elaborated to build a complete application is _____
- A) Subsystem B) Pattern
C) Model D) Framework
17. The primary goal when estimating system performance during the design phase is
- A) To identify the hardware requirements for the system. B) To predict how well the system will perform under different conditions
C) To ensure that all subsystems are fully functional. D) To determine the user interface design requirements.
18. Organizing the system into subsystems helps to
- A) Reduce the overall complexity of the system. B) Focus on user interface design.
C) Minimize the number of hardware components. D) Maximize code duplication across components
19. Choosing a software control strategy during system design is important because
- A) It determines how the system will interact with external hardware. B) It ensures that the system's performance is optimized.
C) It defines the layout of user interface screens. D) It specifies how the software will manage and control its internal operations.
20. Making a reuse plan an important decision during system design because
- A) It allows for the creation of custom-built components. B) It eliminates the need for subsystem organization.
C) It ensures that existing solutions are incorporated, reducing development time and cost. D) It makes it easier to design interfaces between subsystems.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the characteristics of object oriented approach? Briefly explain the stages of object oriented methodology.	8	L*2	1	1
	b) Design and Explain the class model of windowing system.	8	L2	1	1
2.	a) Write a note on the following . (i)Association Class (ii)Aggregation (iii)Multiplicity	8	L2	1	1
	b) What are the uses of Enumeration in modeling ?Differentiate between Enumeration and Generalization with suitable modeling example.	8	L2	2	1
3.	a) Define Event. Explain the different kinds of events with an example.	8	L2	2	1
	b) What is Multiple inheritance? Explain the kinds of multiple inheritance with suitable illustration.	8	L2	2	1

Unit – II

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| 4. a) | How nested state diagram overcome the problems of state diagram? Design and explain the nested state diagram for Vending Machine. | 8 | L3 | 3 | 1 |
| b) | Illustrate the concept of aggregation Concurrency model with an example of car transmission system. | 8 | L2 | 3 | 1 |
| 5. a) | What is Sequence Model? Design and briefly discuss the sequence diagram for the following scenario.
(i) To establish a session with an online stock broker
(ii) For stock purchase and stock quote
(iii) For stock purchase that fails | 8 | L3 | 3 | 1 |
| b) | What is need of activity diagram? Draw and explain the activity diagram for ATM card verification system. | 8 | L3 | 4 | 1 |
| 6. a) | Briefly Describe the steps performed in application interaction model . | 8 | L2 | 4 | 1 |
| b) | With neat diagram demonstrate the refined ATM Class model having various attribute. | 8 | L2 | 4 | 1 |

Unit – III

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| 7. a) | Briefly discuss various ways of choosing the software strategy. | 8 | L2 | 5 | 1 |
| b) | Differentiate between batch transformation and continuous transformation system. | 8 | L2 | 5 | 1 |
| 8. a) | Illustrate the various steps involved in designing an algorithm. | 8 | L3 | 5 | 1 |
| b) | Elaborate on various tasks involved in design optimization. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
